Chapter 1 The atom

EXTRA WORKSHEET 1:

Ionic compounds—chemical names and formulas

SCIENCE UNDERSTANDING **VERBAL/LINGUISTIC**

Name:_____Class:_____Date: _____

Ionic compounds are made up of a lattice of cations (positive ions) and anions (negative ions). Some common cations and anions are listed in the table below.

Cations			Anions		
Charge	**Name**	**Symbol**	**Name**	**Charge**	
+1	Hydrogen ion	H^+	F^-	Fluoride ion	-1
	Lithium ion	Li^+	Cl^-	Chloride ion	
	Sodium Ion	Na^+	Br^-	Bromide ion	
	Potassium ion	K^+	I^-	Iodide ion	
	Copper (I) ion	Cu^+	O^{2-}	Oxide	-2
+2	Beryllium ion	Be^{2+}	S^{2-}	Sulfide	
	Magnesium ion	Mg^{2+}	N^{3-}	Nitride	-3
	Calcium ion	Ca^{2+}	P^{3-}	Phosphide	
	Manganese (II) ion	Mn^{2+}	C^{4-}	Carbide	-4
	Iron (II) ion	Fe^{2+}			
	Nickel ion	Ni^{2+}			
	Copper (II) ion	Cu^{2+}			
	Zinc ion	Zn^{2+}			
	Strontium ion	Sr^{2+}			
	Barium ion	Ba^{2+}			
	Lead ion	Pb^{2+}			
+3	Boron ion	B^{3+}			
	Aluminium ion	Al^{3+}			
	Chromium (III) ion	Cr^{3+}			
	Iron (III) ion	Fe^{3+}			
+4	Chromium (IV) ion	Cr^{4+}			
	Manganese (IV) ion	Mn^{4+}			
+6	Chromium (VI) ion	Cr^{6+}			

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Naming ionic compounds

The name of an ionic compound is simply the name of the cation combined with the name of the anion. For example, the name for the compound made from magnesium cations (Mg^{2+}) and oxide anions (O^{2-}) is magnesium oxide.

However, some metallic atoms can produce several cations with different charges. For example, copper atoms can form copper cations with charges of +1 or +2. To distinguish between these ions, the charge of the ion is included in the ion name using Roman numerals: the two types of copper ions are named copper (I) ion (Cu^{+}) and copper (II) ion (Cu^{2+}). The Roman numerals are also included in the name of the compound. When copper (I) ions form an ionic compound with chloride ions (Cl^{-}), the compound is named copper (I) chloride. Similarly, when copper (II) ions form an ionic compound with chloride ions (Cl^{-}) it is named copper (II) chloride.

Chemical formulas

The chemical formula for an ionic compound combines the chemical symbols for each ion in the compound, but you don't include the charges on the ions. Sodium chloride, made from sodium ions (Na^{+}) and chloride ions (Cl^{-}), has the chemical formula NaCl.

However, you must ensure that the total charge of the cations and anions in the crystal lattice adds to zero. For example, the chemical formula for magnesium chloride is $MgCl_2$. This is because the magnesium ion (Mg^{2+}) has a charge of +2 and the chloride ion (Cl^{-}) has a charge of -1, so each magnesium ion needs two chloride ions to balance the total charge.

Similarly, the formula for sodium nitride is Na_3N. The sodium ion (Na^{+}) has a charge of +1 and the nitride ion (N^{3-}) has a charge of -3, so there must be three sodium ions to balance the charge of each nitride ion in the crystal lattice.

If the charge of one ion does not divide evenly into the charge of the other, then both ions will need to be multiplied in the chemical formula. For example, the formula for iron (III) oxide is Fe_2O_3. The charge of the iron (III) ion (Fe^{3+}) does not divide evenly

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into the charge of the oxide ion (O^{2-}), so to balance the charge there must be two iron (III) ions for every three oxide ions in the crystal lattice.

1 From the table above, **state** whether anions and cations are generally metallic or non-metallic.

a anions _____

b cations _____

2 **Identify** a non-metallic cation.

_____

3 **List** the names and symbols of all ions in the table above that come from the same atom but have different charges.

_____

_____

_____

_____

4 **Determine** the chemical names of the following ionic compounds from their chemical formulas.

	Chemical formula	Chemical name
a	NaCl	
b	PbI_2	
c	BaS	
d	Li_3N	
e	Al_2O_3	

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5 **Determine** the chemical names of the following ionic compounds from their chemical formulas. Be careful to identify the correct charge of each cation.

	Chemical formula	Chemical name
a	$CuCl_2$	
b	FeO	
c	Fe_2O_3	
d	MnS_2	
e	CrO_3	

6 **Determine** the chemical name and formula of the ionic compounds made from the following cations and anions.

	Cation	Anion	Chemical name	Chemical formula
a	Na^{+}	O^{2-}		
b	Mg^{2+}	Cl^{-}		
c	Zn^{2+}	S^{2-}		
d	Al^{3+}	P^{3-}		
e	Al^{3+}	C^{4-}		
f	Cu^{2+}	S^{2-}		
g	Mn^{4+}	O^{2-}		
h	Cr^{6+}	Cl^{-}		
i	Fe^{2+}	S^{2-}		
j	Fe^{3+}	S^{2-}		

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A solution of sodium sulfide is mixed with a solution of zinc chloride. When these two solutions are mixed, they undergo a chemical reaction to form solid zinc sulfide and a solution of sodium chloride.

7 **Determine** the chemical names and formulas of the ions that are in solution before the chemical reaction.

a_____

b_____

c_____

d_____

8 **Determine** the chemical formula of the solid (precipitate) that is formed by the chemical reaction.

_____

9 **Determine** the chemical names and formulas of the ions that are left in solution after the chemical reaction.

a_____

b_____

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